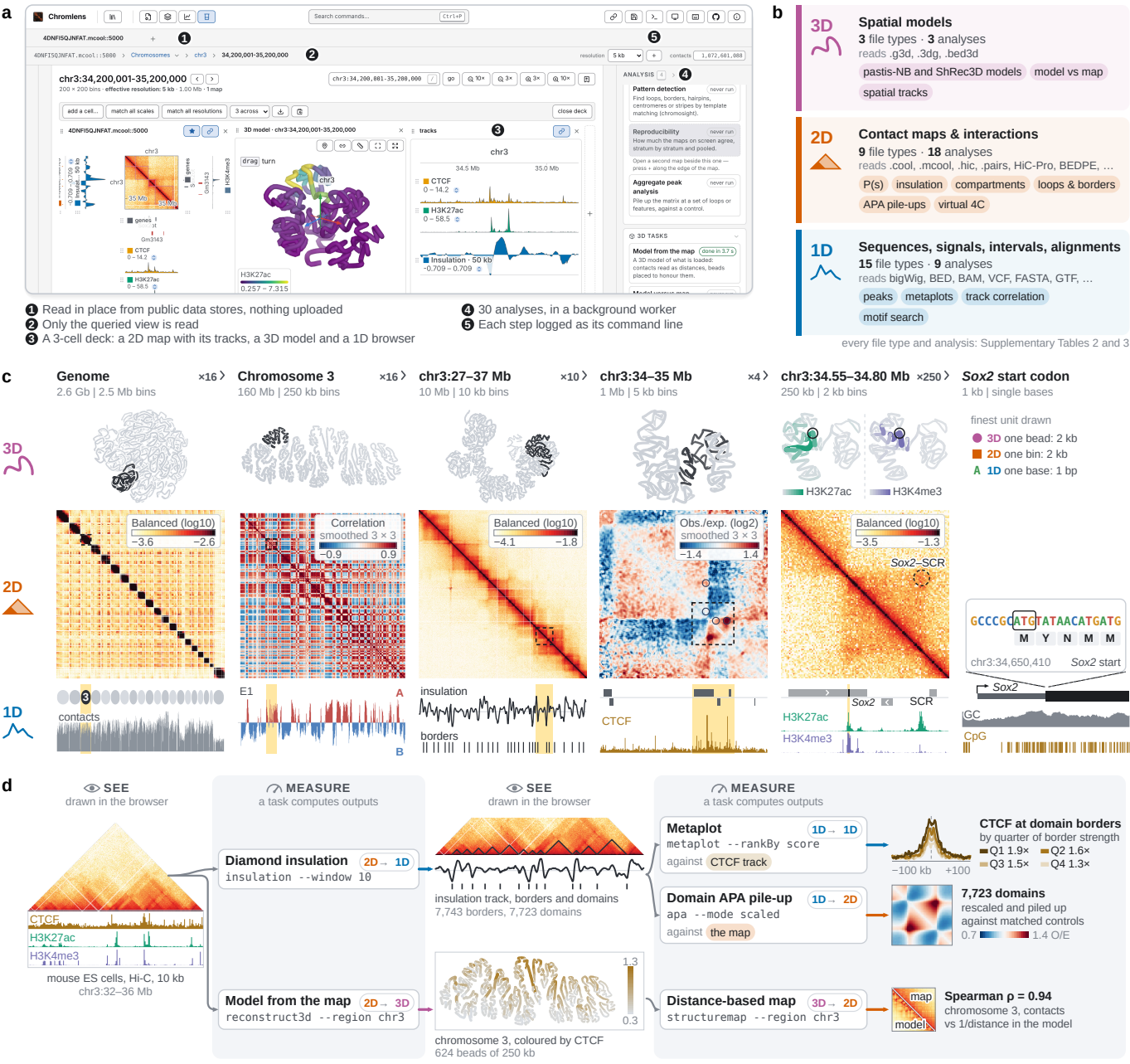

a
Chromlens
Search commands...
Ctrl+P
4DNFISQJNFAT.mcool::5000
4DNFISQJNFAT.mcool::5000 › Chromosomes › chr3 › 34,200,001-35,200,000
resolution 5 kb
contacts 1,072,601,088
chr3:34,200,001-35,200,000
200 × 200 bins · effective resolution: 5 kb · 1.00 Mb · 1 map
chr3:34,200,001-35,200,000
go
10×
3×
3×
10×
add a cell...
match all scales
match all resolutions
3 across
close deck
4DNFISQJNFAT.mcool::5000
3D model · chr3:34,200,001-35,200,000
tracks
drag turn
chr3
34.5 Mb
35.0 Mb
CTCF
0 – 14.2
H3K27ac
0 – 58.5
Insulation · 50 kb
-0.709 – 0.709
H3K27ac
0.257 – 7.315
ANALYSIS
Pattern detection
never run
Find loops, borders, hairpins, centromeres or stripes by template matching (chromosight).
Reproducibility
never run
How much the maps on screen agree, stratum by stratum and pooled.
Open a second map beside this one — press + along the edge of the map.
Aggregate peak analysis
never run
Pile up the matrix at a set of loops or features, against a control.
3D TASKS
Model from the map
done in 3.7 s
A 3D model of what is loaded: contacts read as distances, beads placed to honour them.
1 Read in place from public data stores, nothing uploaded
2 Only the queried view is read
3 A 3-cell deck: a 2D map with its tracks, a 3D model and a 1D browser
4 30 analyses, in a background worker
5 Each step logged as its command line
b
3D
Spatial models
3 file types · 3 analyses
reads .g3d, .3dg, .bed3d
pastis-NB and ShRec3D models
model vs map
spatial tracks
2D
Contact maps & interactions
9 file types · 18 analyses
reads .cool, .mcool, .hic, .pairs, HiC-Pro, BEDPE, …
P(s)
insulation
compartments
loops & borders
APA pile-ups
virtual 4C
1D
Sequences, signals, intervals, alignments
15 file types · 9 analyses
reads bigWig, BED, BAM, VCF, FASTA, GTF, …
peaks
metaplots
track correlation
motif search
every file type and analysis: Supplementary Tables 2 and 3
c
Genome
2.6 Gb | 2.5 Mb bins
×16
Chromosome 3
160 Mb | 250 kb bins
×16
chr3:27–37 Mb
10 Mb | 10 kb bins
×10
chr3:34–35 Mb
1 Mb | 5 kb bins
×4
chr3:34.55–34.80 Mb
250 kb | 2 kb bins
×250
Sox2 start codon
1 kb | single bases
3D
2D
1D
finest unit drawn
3D one bead: 2 kb
2D one bin: 2 kb
1D one base: 1 bp
H3K27ac
H3K4me3
Balanced (log10)
−3.6
−2.6
Correlation
smoothed 3 × 3
−0.9
0.9
Balanced (log10)
−4.1
−1.8
Obs./exp. (log2)
smoothed 3 × 3
−1.4
1.4
Balanced (log10)
−3.5
−1.3
Sox2–SCR
GCCCGCATGTATAACATGATG
M Y N M M
chr3:34,650,410
Sox2 start
Sox2
3
contacts
E1
A
B
insulation
borders
CTCF
Sox2
SCR
H3K27ac
H3K4me3
GC
CpG
d
SEE
drawn in the browser
MEASURE
a task computes outputs
SEE
drawn in the browser
MEASURE
a task computes outputs
CTCF
H3K27ac
H3K4me3
mouse ES cells, Hi-C, 10 kb
chr3:32–36 Mb
Diamond insulation
2D → 1D
insulation --window 10
insulation track, borders and domains
7,743 borders, 7,723 domains
Model from the map
2D → 3D
reconstruct3d --region chr3
1.3
0.3
chromosome 3, coloured by CTCF
624 beads of 250 kb
Metaplot
1D → 1D
metaplot --rankBy score
against CTCF track
−100 kb
+100
CTCF at domain borders
by quarter of border strength
Q1 1.9×
Q2 1.6×
Q3 1.5×
Q4 1.3×
Domain APA pile-up
1D → 2D
apa --mode scaled
against the map
7,723 domains
rescaled and piled up against matched controls
0.7
1.4 O/E
Distance-based map
3D → 2D
structuremap --region chr3
map
model
Spearman ρ = 0.94
chromosome 3, contacts vs 1/distance in the model